Classwork 03//Computational Thinking, Convolutional Neural Net

Student Name: ______________________ **Points:** 30

A Number: ______________________ **Due:** Feb 17, 2026

1 Log (1+x) Layer (10 Points)

Provide a code-snippet of how would you design a neural network Layer using PyTorch by inheriting from `nn.Module` that performs $\log(1+x)$ transformation on the feature matrix.

1.1 Solution

```
class LogLayer(nn.Module):
    def __init__(self):
        super().__init__()

    def forward(self, x):
        logx = torch.log(x + 1)
        return logx

model = LogLayer()
```

2 Convolution Dimension (5 Points)

Consider a kernel of dimension 3×3 operating on an input image dimension of 5×5 with a stride of 1. Consider no padding, what is the dimension of the output??

2.1 Solution

3×3.

3 Convolution Computation (10 Points)

Given input matrix as

$$I = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 1 \\ 7 & 8 & 9 & 2 \\ 0 & 1 & 2 & 3 \end{bmatrix}$$

and kernel as

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Calculate the output feature map using convolution with a stride of 1.

3.1 Solution

$$Output = \begin{bmatrix} -6 & 12 \\ -6 & 8 \end{bmatrix}$$

4 Max Pooling (5 Points)

Apply max pooling with a stride of 2 to

$$I = \begin{bmatrix} 3 & 2 & 5 & 1 \\ 4 & 1 & 6 & 2 \\ 1 & 7 & 3 & 4 \\ 2 & 5 & 1 & 3 \end{bmatrix}$$

4.1 Solution

$$O = \begin{bmatrix} 4 & 6 \\ 7 & 4 \end{bmatrix}$$